Derivatives: Contracts, Models, and Markets · Technical Note 31

Fitting, Option Volatilities, and the Second Factor

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Chapter 31 of Derivatives: Contracts, Models, and Markets states four results and derives none of them, because each needs a page of stochastic calculus that would break the chapter's line of argument. This note supplies them. Part I answers the question the no-arbitrage generation rests on: why *a drift chosen as a function of time can reproduce any initial curve at all, and not merely approximately. The answer is that the fitting requirement, once differentiated in maturity, collapses to a first-order linear equation in one unknown function—so the correspondence between the drift and the curve is one-to-one, explicit, and triangular, which is also why the tree's displacement is solved rather than iterated. Part II derives equation (31.8), the forward bond volatility that makes the chapter's European formulas exact rather than approximate, and shows where mean reversion puts a ceiling on it. Part III collects the G2++ analytics the chapter asserts: the variance functional, the shift that fits the curve, the bond price, and the two-factor volatility that turns the one-factor option formulas into two-factor ones unchanged—together with the reason Jamshidian's decomposition does not survive the second factor. Part IV builds the two-dimensional tree, and finds the obstruction the chapter alludes to: the natural correlation correction to the nine branching probabilities is admissible only for $|\rho| \le 1/3$, which typical two-factor calibrations violate. The repair is a change of variable that moves the correlation out of the probabilities and into the drift, where the machinery of Section 31.8 already handles it.*

Every result here is recomputed by `tools/verify/tn31-fitting-and-two-factors.py`.

Part I. Why a fitted drift can reproduce any curve

1 THE QUESTION

Section 31.6's model is

$$dr = \big[\theta(t) - ar\big]dt + \sigma\, dW \tag{TN31.1}$$

and the chapter asserts that θ can be chosen so that the model's zero-coupon curve equals today's, exactly, for every maturity at once. That is a strong claim. The curve has as many points as there are maturities; the model has one function of time. The claim is that this is enough, and the reason is worth seeing rather than accepting.

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2 THE BOND PRICE IN CLOSED FORM

Solve (TN31.1) by the usual integrating factor:

$$r(t) = r(0)e^{-at} + \int_0^t e^{-a(t-u)}\theta(u)\,du + \sigma\int_0^t e^{-a(t-u)}\,dW(u) \qquad \text{(TN31.2)}$$

so $r(t)$ is Gaussian with mean

$$m(t) = r(0)e^{-at} + \int_0^t e^{-a(t-u)}\theta(u)\,du$$

and a variance that does not involve θ at all. Write $I = \int_0^T r(s)\,ds$. A linear functional of a Gaussian process is Gaussian, so I is Gaussian, and

$$P(0,T) = \hat{E}\left[e^{-I}\right] = \exp\left(-\hat{E}[I] + \frac{1}{2}\operatorname{Var}(I)\right) \qquad \text{(TN31.3)}$$

This is the step that makes the whole construction work, and it is available only because the model is Gaussian: the expectation of the exponential is the exponential of the mean and variance, with nothing left over.

The two moments are explicit. The mean is $\hat{E}[I] = \int_0^T m(s)\,ds$. For the variance, exchange the order of integration in the stochastic term of (TN31.2):

$$\int_0^T\left[\sigma\int_0^s e^{-a(s-u)}dW(u)\right]ds = \sigma\int_0^T\left[\int_u^T e^{-a(s-u)}ds\right]dW(u) = \frac{\sigma}{a}\int_0^T\left(1-e^{-a(T-u)}\right)dW(u)$$

and Itô's isometry gives

$$\operatorname{Var}(I) = \frac{\sigma^2}{a^2}\left[T - \frac{2\left(1-e^{-aT}\right)}{a} + \frac{1-e^{-2aT}}{2a}\right] \qquad \text{(TN31.4)}$$

Note what is *not* in (TN31.4): θ. The drift moves the mean of I and nothing else, which is the structural fact the rest of Part I exploits.

3 THE FITTING CONDITION, AND WHY IT HAS A UNIQUE SOLUTION

Let $P^M(0,T)$ be today's observed curve and $f^M(0,T) = -\partial \ln P^M(0,T)/\partial T$ its forward curve. Requiring the model's $P(0,T)$ in (TN31.3) to equal $P^M(0,T)$ for every T, and taking logarithms, gives

$$\int_0^T m(s)\,ds - \frac{1}{2}\operatorname{Var}(I) = \int_0^T f^M(0,u)\,du$$

This is one equation for each maturity, and an integral equation looks harder than it is. *Differentiate in T.* Differentiating (TN31.4) gives

$$\frac{\partial}{\partial T}\operatorname{Var}(I) = \frac{\sigma^2}{a^2}\left[1 - 2e^{-aT} + e^{-2aT}\right] = \frac{\sigma^2}{a^2}\left(1-e^{-aT}\right)^2$$

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so the condition becomes the pointwise statement

$$m(T) = f^M(0,T) + \frac{\sigma^2}{2a^2}\left(1 - e^{-aT}\right)^2 \tag{TN31.5}$$

That is the whole answer. The requirement on the curve, which looked like infinitely many simultaneous conditions, is a formula for the mean of the short rate at each date—one number per date, determined by the curve at that date and nothing else. The second term is the price of the exponential in (TN31.3): the risk-neutral mean of $r(T)$ is *above* the forward rate, by a convexity term that grows to $\sigma^2/2a^2$.

And m determines θ. Differentiating the definition of m gives $m'(t) = \theta(t) - a\,m(t)$, so

$$\theta(t) = m'(t) + a\,m(t)$$

Substituting (TN31.5) and collecting the exponentials,

$$m'(t) = \frac{\partial f^M(0,t)}{\partial t} + \frac{\sigma^2}{a}\left(e^{-at} - e^{-2at}\right), \qquad a\,m(t) = a f^M(0,t) + \frac{\sigma^2}{2a}\left(1 - e^{-at}\right)^2$$

and the exponential terms combine to $\sigma^2(1 - e^{-2at})/2a$, leaving

$$\theta(t) = \frac{\partial f^M(0,t)}{\partial t} + a f^M(0,t) + \frac{\sigma^2}{2a}\left(1 - e^{-2at}\right) \tag{TN31.6}$$

which is the formula Section 31.6 quotes. Letting $a \to 0$ sends the last term to $\sigma^2 t$ and the middle one to zero, returning Section 31.5's Ho–Lee drift, as it must.

Three things are now clear that were not before. The fit is exact, not approximate, because (TN31.5) is an identity rather than a minimization. It is unique, because a first-order linear equation with a given initial condition has one solution. And it requires the curve to be differentiable twice, since (TN31.6) contains $\partial f/\partial t$—which is why a curve built by linear interpolation on forward rates produces a θ with jumps, and why production curve builders use methods that are at least once continuously differentiable in the forwards.

4 WHY THE TREE'S DISPLACEMENT IS SOLVED RATHER THAN FITTED

Section 31.8 builds a tree whose displacement $\alpha(i)$ makes each level reprice one discount factor, and the chapter says the curve comes back to machine precision because $\alpha(i)$ solves one equation in one unknown. Part I explains why that structure is available at all: the same triangularity. In continuous time, θ on $[0,T]$ determines $P(0,T)$ and no later value of θ affects it; on the lattice, $\alpha(0),\ldots,\alpha(i)$ determine the level-i Arrow–Debreu prices and therefore $P(0,(i+1)\Delta t)$, while $\alpha(i+1)$ and beyond are still free. *A triangular system is solved forwards, one unknown at a time, and never needs an optimizer.* The exactness of the tree's fit is the discrete image of the exactness of (TN31.5).

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Part II. Equation (31.8), the forward bond volatility

5 THE BOND'S VOLATILITY

Equation (31.2) makes $\ln P(t,T)$ affine in r:

$$\ln P(t,T) = \ln A(t,T) - B(t,T)\, r(t), \qquad B(t,T) = \frac{1 - e^{-a(T-t)}}{a}$$

Only r is stochastic and B is deterministic, so the diffusion term of $d \ln P(t,T)$ is $-B(t,T)\sigma\, dW$. The T-maturity bond therefore has volatility $\sigma B(t,T)$, growing with maturity but—because B is bounded by $1/a$—never beyond σ/a. *Mean reversion is already visible at this stage*, before any option appears.

6 THE FORWARD BOND PRICE, AND WHY ITS VOLATILITY IS A NUMBER

A European option expiring at T on the S-maturity bond, $S > T$, is an option on the forward bond price

$$F(t) = \frac{P(t,S)}{P(t,T)}$$

which is a martingale under the T-forward measure. Subtracting the two log-dynamics, its diffusion coefficient is

$$\nu(t) = \sigma\,[B(t,S) - B(t,T)] = \frac{\sigma}{a}\left[e^{-a(T-t)} - e^{-a(S-t)}\right] = \frac{\sigma}{a}\left(1 - e^{-a(S-T)}\right) e^{-a(T-t)}$$

The factorization in the last step is the point. *The maturity gap $S - T$ and the time to expiration $T - t$ separate*, and neither factor contains r. So ν is a deterministic function of time, and the accumulated variance over the option's life is an ordinary integral:

$$\int_0^T \nu(t)^2\, dt = \frac{\sigma^2}{a^2}\left(1 - e^{-a(S-T)}\right)^2 \int_0^T e^{-2a(T-t)}\, dt = \frac{\sigma^2}{a^2}\left(1 - e^{-a(S-T)}\right)^2 \frac{1 - e^{-2aT}}{2a}$$

Taking the square root gives equation (31.8),

$$\sigma_P(T,S) = \frac{\sigma}{a}\left(1 - e^{-a(S-T)}\right)\sqrt{\frac{1 - e^{-2aT}}{2a}} \tag{TN31.7}$$

A lognormal variable whose volatility is deterministic is exactly Black's assumption, so the chapter's (31.9) and (31.10) are exact prices in this model and not approximations—which is the claim Section 31.7 makes and the reason it can afford to.

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7 WHERE THE CEILING COMES FROM

Both factors in (TN31.7) saturate, and they saturate for different reasons. The first, $1 - e^{-a(S-T)}$, rises to one as the underlying bond lengthens: beyond a maturity gap of a few multiples of $1/a$, a longer bond is no more volatile. The second, $\sqrt{(1 - e^{-2aT})/2a}$, rises to $1/\sqrt{2a}$ as the option lengthens. So

$$\sigma_P(T, S) \longrightarrow \frac{\sigma}{a\sqrt{2a}} \qquad \text{as both } T \text{ and } S - T \text{ grow}$$

At $a = 0.08$ and $\sigma = 1\%$ that ceiling is 31.3%, and at a two-year expiration the limit over S is already 16.4%. This is the mechanism behind the observation in Example 31.4 that the caplets rise in value with expiration although their forward rate is identical: σ_P is what differs, and its growth is bounded. *Without mean reversion the second factor would be $\sqrt{T}$ without limit*, which is Ho–Lee, and is the defect Section 31.5 rules the model out for.

Part III. The two-factor analytics

8 THE MODEL AND ITS VARIANCE FUNCTIONAL

Write the G2++ form directly, as Section 31.13 recommends:

$$r(t) = x(t) + y(t) + \varphi(t), \qquad \begin{aligned} dx &= -a\,x\,dt + \sigma\,dW_1, & x(0) &= 0 \\ dy &= -b\,y\,dt + \eta\,dW_2, & y(0) &= 0 \end{aligned} \tag{TN31.8}$$

with $dW_1\,dW_2 = \rho\,dt$. Both factors start at zero and revert to zero; the entire curve level lives in φ.

Everything follows from one quantity, the variance of the integrated short rate. Repeating Part I's exchange of integration order for each factor, and adding the cross term,

$$\begin{aligned} V(t,T) = {} & \frac{\sigma^2}{a^2}\left[\tau - \frac{2(1 - e^{-a\tau})}{a} + \frac{1 - e^{-2a\tau}}{2a}\right] + \frac{\eta^2}{b^2}\left[\tau - \frac{2(1 - e^{-b\tau})}{b} + \frac{1 - e^{-2b\tau}}{2b}\right] \\ & + \frac{2\rho\sigma\eta}{ab}\left[\tau - \frac{1 - e^{-a\tau}}{a} - \frac{1 - e^{-b\tau}}{b} + \frac{1 - e^{-(a+b)\tau}}{a+b}\right], \qquad \tau = T - t \end{aligned} \tag{TN31.9}$$

The first two brackets are (TN31.4) for each factor. The third is the only new algebra, and its shape—three reverting terms and one at the combined rate $a + b$—is what the second factor costs.

9 THE SHIFT, AND THE BOND PRICE

The fitting argument of Part I goes through unchanged, with φ in place of θ and one less differentiation, because φ enters r directly rather than through a drift. Requiring

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$\exp(-\int_0^T \varphi + V(0,T)/2) = P^M(0,T)$ and differentiating,

$$\varphi(T) = f^M(0,T) + \frac{\sigma^2}{2a^2}\left(1-e^{-aT}\right)^2 + \frac{\eta^2}{2b^2}\left(1-e^{-bT}\right)^2 + \frac{\rho\sigma\eta}{ab}\left(1-e^{-aT}\right)\left(1-e^{-bT}\right) \tag{TN31.10}$$

the two-factor twin of (TN31.5), with a cross term that is negative whenever ρ is—which is the usual calibrated sign.

The bond price is then, for $\tau = T - t$ and $B_a(t,T) = (1-e^{-a\tau})/a$,

$$P(t,T) = \frac{P^M(0,T)}{P^M(0,t)}\exp\left\{\frac{1}{2}\Big[V(t,T) - V(0,T) + V(0,t)\Big] - B_a(t,T)\,x(t) - B_b(t,T)\,y(t)\right\} \tag{TN31.11}$$

The shift has disappeared from it, exactly as θ disappears from the one-factor $\ln A(t,T)$ of Section 31.6, and for the same reason: φ was solved for the curve, so the curve can be carried in its place. At $t = 0$ the exponent is zero and (TN31.11) returns $P^M(0,T)$ identically, which is the cheapest check on any implementation.

10 EUROPEAN OPTIONS: THE ONE-FACTOR FORMULAS, UNCHANGED

Since $\ln P(T,S)$ is affine in the jointly Gaussian pair $(x(T), y(T))$, it is Gaussian, and the forward bond price is lognormal with deterministic volatility—Part II's argument again, now with two drivers. The accumulated variance is

$$\begin{aligned}\Sigma(T,S)^2 = {} & \frac{\sigma^2}{2a^3}\left(1-e^{-a(S-T)}\right)^2\left(1-e^{-2aT}\right) + \frac{\eta^2}{2b^3}\left(1-e^{-b(S-T)}\right)^2\left(1-e^{-2bT}\right) \\ & + \frac{2\rho\sigma\eta}{ab(a+b)}\left(1-e^{-a(S-T)}\right)\left(1-e^{-b(S-T)}\right)\left(1-e^{-(a+b)T}\right)\end{aligned} \tag{TN31.12}$$

and the bond call and put are Section 31.7's (31.9) and (31.10) verbatim, with Σ in place of σ_P:

$$\text{ZBC} = P(0,S)\Phi(h) - XP(0,T)\Phi(h-\Sigma), \qquad h = \frac{1}{\Sigma}\ln\frac{P(0,S)}{X\,P(0,T)} + \frac{\Sigma}{2}$$

Setting $\eta = 0$ collapses (TN31.12) to σ_P^2, since the first term is (TN31.7) squared. *So the second factor costs nothing at all on a European option on a single bond*—it changes one number.

11 WHERE JAMSHIDIAN STOPS WORKING

It costs a great deal on a swaption. Section 31.7's decomposition replaces an option on a coupon bond by a portfolio of options on the zeros, and it needs the coupon bond's value to be a monotone function of *one* state variable, so that the exercise region is an interval and the threshold rate r^* exists. Under (TN31.8) the coupon bond depends on x and y through two different maturity weightings, B_a and B_b. The exercise boundary is a curve in the (x,y) plane, not a point, and there is no r^* to find.

This is the real cost of the second factor, and it is why Section 31.13 says two-factor implementations price by simulation or by a two-dimensional grid. The standard

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salvage is to condition on one factor and integrate: for each value of $x(T)$ the remaining problem is one-dimensional in $y(T)$ and Jamshidian applies to it, so the swaption becomes a one-dimensional numerical integral of closed forms. That is fast and exact, but it is no longer a formula.

Part IV. The two-dimensional tree

12 THE GEOMETRY

Build a grid for each factor exactly as Section 31.8 does, since each of x and y is the zero-drift mean-reverting process (31.12) that section treats:

$$\Delta x = \sigma\sqrt{3\Delta t}, \qquad \Delta y = \eta\sqrt{3\Delta t},$$

with the truncation levels from that section's rule applied to each mean reversion separately—$j_{\max}$ the smallest integer at or above $0.1835/(a\Delta t)$, and $k_{\max}$ likewise with b. *The two limits are usually very different*, because a two-factor calibration wants one fast factor and one slow one: at $a = 0.50$, $b = 0.05$ and $\Delta t = 0.25$, $j_{\max} = 2$ while $k_{\max} = 15$. The grid is a narrow band in the fast factor and a wide one in the slow factor, which is the shape to expect and a useful check on an implementation.

A node is now a pair (j, k) and branches to nine successors. The marginal probabilities in each direction are Section 31.8's, including the switched patterns at $\pm j_{\max}$ and $\pm k_{\max}$; what is left is to install the correlation.

13 THE CORRELATION, AND THE OBSTRUCTION

Let p^x and p^y be the two marginal triples and o^x, o^y their offsets. The product $p_{jk} = p^x_j p^y_k$ matches both marginals and produces zero covariance, so write

$$p_{jk} = p^x_j\, p^y_k + \varepsilon_{jk}$$

and require three things of ε: its rows sum to zero and its columns sum to zero, so the marginals survive; and it delivers the covariance. Over one step $\mathrm{Cov}(\Delta x, \Delta y) = \rho\sigma\eta\,\Delta t$, so in lattice units the requirement is

$$\sum_{j,k} \varepsilon_{jk}\, o^x_j\, o^y_k = \frac{\rho\,\sigma\eta\,\Delta t}{\Delta x\,\Delta y} = \frac{\rho}{3}$$

independent of every parameter, because the spacings were chosen as $\sigma\sqrt{3\Delta t}$ and $\eta\sqrt{3\Delta t}$. The simplest ε meeting the two sum conditions puts $\pm e$ at the four corners,

$$\varepsilon = e\begin{pmatrix} +1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & +1 \end{pmatrix}, \qquad \sum \varepsilon\, o^x o^y = 4e \quad\Longrightarrow\quad e = \frac{\rho}{12}$$

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Now check positivity at the center, where the marginals are $(1/6, 2/3, 1/6)$ and the corner probabilities are $1/36$. The cell that the correction reduces carries

$$\frac{1}{36} - \frac{|\rho|}{12} = \frac{1 - 3|\rho|}{36} \;\geq\; 0 \qquad \Longleftrightarrow \qquad |\rho| \leq \frac{1}{3}$$

And that is a real obstruction, not a technicality. A calibrated G2++ typically has ρ between -0.7 and -0.9, since the second factor is there precisely to let the short and long ends move apart. At $\rho = -0.70$ the corner probability is -0.031: negative, and the tree is not a tree. Spreading ε over the edge cells as well buys some room—the zero-row-and-column-sum matrices form a four-dimensional space, so there is freedom to solve for an admissible correction node by node—but the freedom runs out too, and at the boundary nodes, where Section 31.8's switched branching has already skewed the marginals, it runs out sooner.

14 THE REPAIR: MOVE THE CORRELATION INTO THE DRIFT

The way out is to change variables so that the two drivers are independent, and it costs nothing. Write $dW_2 = \rho\, dW_1 + \sqrt{1 - \rho^2}\, dW_3$ with W_3 independent of W_1, and set

$$\tilde{y} = y - c\, x, \qquad c = \frac{\eta\rho}{\sigma}$$

The constant c is chosen to cancel the dW_1 loading exactly, and

$$d\tilde{y} = \Big[-b\, \tilde{y} + c\, (a - b)\, x\Big] dt + \eta\sqrt{1 - \rho^2}\, dW_3 \tag{TN31.13}$$

with $r = (1 + c)\, x + \tilde{y} + \varphi$. *The diffusions are now independent*, so the nine probabilities are a plain product with no correction and no admissibility question, at any ρ. The correlation has not gone away: it has moved into the drift of $\tilde{y}$, where it appears as a term proportional to x.

And a drift is what Section 31.8's apparatus is built to absorb. That section already handles a state-dependent drift by choosing which node to branch to—the whole point of $j_{\max}$ and the switched patterns is that the drift is matched by *node placement* rather than by probabilities. Here the same device applies in the $\tilde{y}$ direction, with the branching center at level k displaced according to the current x. When $a = b$ the coupling term vanishes and the two trees are exactly independent; when a and b are close, as a slow-factor pair sometimes is, the displacement is small.

The sequence to implement, then, is the one Section 31.8 already describes, run twice. Build the two grids. Take the product probabilities. Displace the $\tilde{y}$ branching by the x-dependent drift in (TN31.13). Then roll the Arrow–Debreu prices forward over the two-dimensional level and solve for the displacement, which is still one equation in one unknown—summing $Q(i, j, k)$ over both indices changes nothing about the triangularity Part I identified, so the two-factor tree reprices the initial curve to machine precision exactly as the one-factor tree does.

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15 WHAT IT COSTS

The node count at level i grows as the product of the two widths rather than their sum, so a tree that was $O(i)$ wide is now $O(i^2)$, and a Bermudan that took milliseconds on the one-factor tree of Example 31.8 takes seconds. That is the arithmetic behind Section 31.13's warning, and it is also why the choice between a two-factor tree and a two-dimensional finite difference grid is usually decided by which is easier to write rather than by speed: they have the same complexity, and the grid has the better convergence.

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